MRDVS



S10 RGBD Camera

Product User Manual

Version： V2.1

Date： 2026-04-22

Revision Record

Version	Date	Instructions
V2.0	2026-03-26	Version 2.0
V2.1	2026-04-01	Updating the wiring sequence instructions

Term interpretation

Term	Interpretation
Depth	Depth video streams are similar to color or black-and-white video streams, where each pixel has a value representing the distance from the camera, that is, 'depth'.
FOV	Field of view, used to describe the range of angles from which a camera images a given scene, mainly includes horizontal field of view (HFOV), vertical field of view (VFOV), and diagonal field of view (DFOV).
IR	Infrared (IR) communication is a wireless communication technology for transmitting data, and this product mainly focuses on near-infrared band imaging
SoC	System-on-chip for processing image data, depth settlement, application algorithm operations, and other functions
dToF	By directly measuring the time interval from the emission to the reception of the light pulse, calculate the distance of the object using the flight time of the light. Light pulses are emitted and received at very short intervals (typically nanoseconds), and the distance is measured by measuring the time difference of the pulses
iToF	The depth information of the target object is obtained by indirectly calculating the light flight time by measuring the phase difference of the modulated light signal
IMU	Inertial measurement unit , a sensor device used to measure the three-axis attitude Angle (or angular rate) and linear acceleration of an object
SLAM	Simultaneous localization and mapping

Content

1 Product Overview 4
1.1 Product Introduction 4
1.2 Functional Features 4
1.2.1 Precise long-distance perception and anti-motion blur 4
1.2.2 Strong anti-multi-path effect capability 5
1.2.3 High reflection object and dynamic reflectivity range adaptability 5
1.2.4 Low power consumption and high response speed 5
1.2.5 Supporting multiple machines to work simultaneously 5
1.2.6 Supporting operation in strong outdoor light conditions 5
1.2.7 Built-in obstacle avoidance algorithm, supporting direct output of signals from I/O ports 6
2 Specification Parameters 6
2.1 Basic Specifications 6
2.2 Interface Definition 7
2.2.1 Camera Interface Description 8
2.2.2 IO Wiring Instructions 8
2.2.3 Cabling Sequence Instructions 8
3 Product Information 9
3.1 Structural Drawings 9
3.2 Camera Components 10
4 Camera Integration Guide 11
4.1 Camera Detection Range 11
4.1.1 3D View of Field of View for S10 Series Cameras 11
4.1.2 2D diagram of field of view for S10 series cameras 11

4.2 (Important) Installation Notes....12
4.2.1 Recommended Installation Methods....12
4.2.2 Incorrect Installation Methods....12
4.2.3 Reasons for camera malfunction....13
4.3 SDK....15
4.4 Operating environment and reliability....15
5 Quick Usage Guide....16
5.1 Packing list....16
5.2 Connection test....16
5.2.1 Camera power supply....16
5.2.2 Network configuration....17
5.2.3 Running the software....17
5.3 Interface Instructions....17
6 Frequently Asked Questions....19
7 Notes....20

1 Product Overview

1.1 Product Introduction

MRDVS S10 is a series of industrial RGB-D cameras with a compact size, high performance and excellent cost-effectiveness. This series features a wide detection range, with the standard version covering 0.3-8 meters and the longest version covering 0.3-17 meters. The products come with SoCs and have spatial and temporal alignment capabilities, and can output depth maps, intensity maps, RGB images, point cloud maps, and obstacle avoidance algorithm results. The cameras not only support Ethernet interfaces but also offer 2 channels of IO output and 2 channels of IO input, allowing for the output of 4 groups of obstacle avoidance area switching and emergency stop signals, deceleration signals, and safety signals.

1.2 Functional Features

1.2.1 Precise long-distance perception and anti-motion blur

Based on dToF technology, the camera can accurately measure distances in dynamic indoor/outdoor environments, providing stable and clear depth data for fast-moving unmanned vehicles, forklifts, drones, and other equipment.

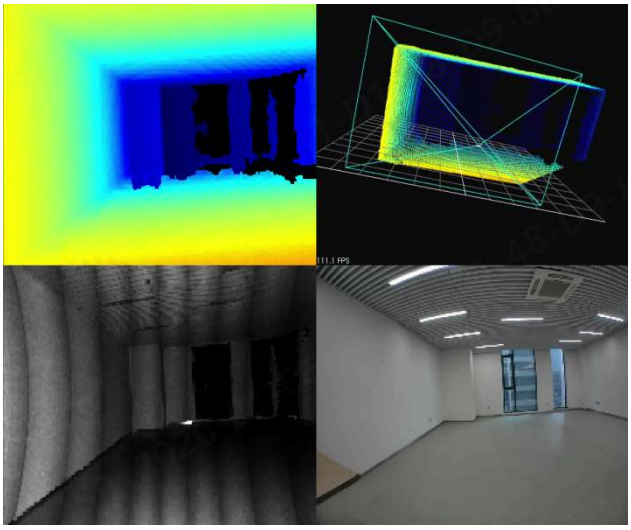


Figure 1 Indoor open space scene: Multi-modal data from the camera (upper left: depth map / lower left: intensity map / upper right: point cloud map / lower right: RGB image)

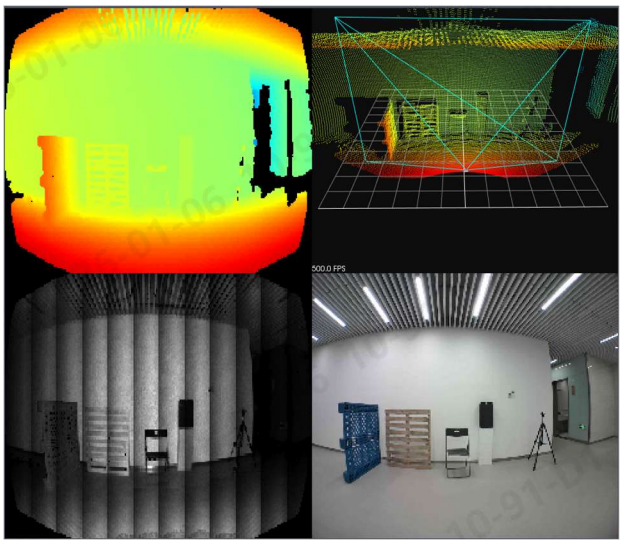


Figure 2 Indoor target scene: Multi-modal data from the camera (upper left: depth map / lower left: intensity map / upper right: point cloud map / lower right: RGB image)

1.2.2 Strong anti-multi-path effect capability

Compared to iToF cameras, in narrow environments, the S10 is better able to handle the overlapping and interference of reflected signals in the environment, providing more reliable distance data.

1.2.3 High reflection object and dynamic reflectivity range adaptability

The S10 can more accurately extract the depth information of high-reflectivity objects, ensuring stability and high performance in high dynamic range scenes.

1.2.4 Low power consumption and high response speed

The camera maintains a low energy consumption level while achieving high response speed (the maximum frame rate can reach 20fps, and for S10PRO, it can reach 10fps), promptly responding to sudden obstacles, providing the robot with efficient and precise obstacle avoidance capabilities.

1.2.5 Supporting multiple machines to work simultaneously

In a multi-robot collaboration environment, S10 has been optimized with a dToF chip, which eliminates interference among multiple robots and enables multiple devices to work simultaneously.

1.2.6 Supporting operation in strong outdoor light conditions

S10 has extremely strong resistance to sunlight. At 100KLUX, for objects with a 90% reflectivity, the S10's distance measurement can reach 8 meters, and the S10 PRO's distance measurement can reach 17 meters.

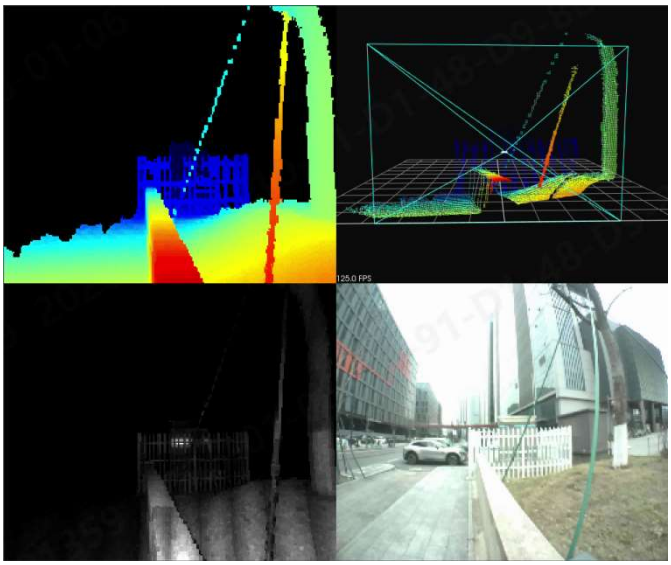


Figure 3 Outdoor lighting scene: Multi-modal data from the camera (upper left: depth map / lower left: intensity map / upper right: point cloud map / lower right: RGB image)

1.2.7 Built-in obstacle avoidance algorithm, supporting direct output of signals from I/O ports

Users can choose between two operation modes: transmitting point cloud data via Ethernet and integrating it into the existing obstacle avoidance system, or directly using the built-in obstacle avoidance algorithm and sending deceleration or stop commands to the controller through I/O ports.

2 Specification Parameters

2.1 Basic Specifications

Product model	**S10 (Lite version is provided)**	**S10Pro**
Laser wavelength	940 nm	940 nm
Output	Depth map/RGB/IR map	Depth map/RGB/IR map
TOF resolution/fps	240×160 @Max. 20fps, typical 15fps	240×160 @Max. 10fps, typical 10fps

TOF perspective	90°×60°±3° 120°×80°±3°	61°×90°±3°
RGB resolution/fps[1]	1920×1080 @Max. 20fps, typical 15fps 1632×1224 @Max. 20fps, typical 15fps	1920×1080 @Max. 10fps, typical 10fps
RGB perspective (H×V)[1]	90°×60°±3°; 120°×80°±3° (The lite version is without RGB)	61°×90°±3°
Detection range	0.3-8m (90% reflectivity)	0.3~17m (90% reflectivity)
Precision	0.3-3m (5% reflectivity)	0.3~13m (10% reflectivity)
Average power consumption	≤3 cm	≤3 cm
Size [2]	≤4w	≤7w
Weight	80 mm×37 mm×25 mm	92 mm×47 mm×51 mm
Power supply	190g	460g
Communication interface	12V~28V	24V
Protection class	Ethernet/GMSL[3]/IO[4]	Ethernet/IO
Eye safety	IP54	IP67
Operating temperature	-20°C~60°C	-20°C~60°C
Storage temperature	-40°C~85°C	-40°C~85°C
Time synchronization	PTP	PTP
Software environment	C/C++/ROS SDK	C/C++/ROS SDK
Supported systems	Windows 7/8/10/11, Linux, Arm Linux/ROS	Windows 7/8/10/11, Linux, Arm Linux/ROS
Anti-light interference	100 KLUX	100KLUX

[1] Offering Lite version without RGB; [2] Please refer to the 3D model for the size; [3] For the GMSL version, please inquire in detail; [4] The Lite 120° version of S10 comes with an IO cable.

2.2 Interface Definition

2.2.1 Camera Interface Description

There are a total of two interfaces on the camera end: the DC 24V power interface and the 100Mbps Ethernet interface.

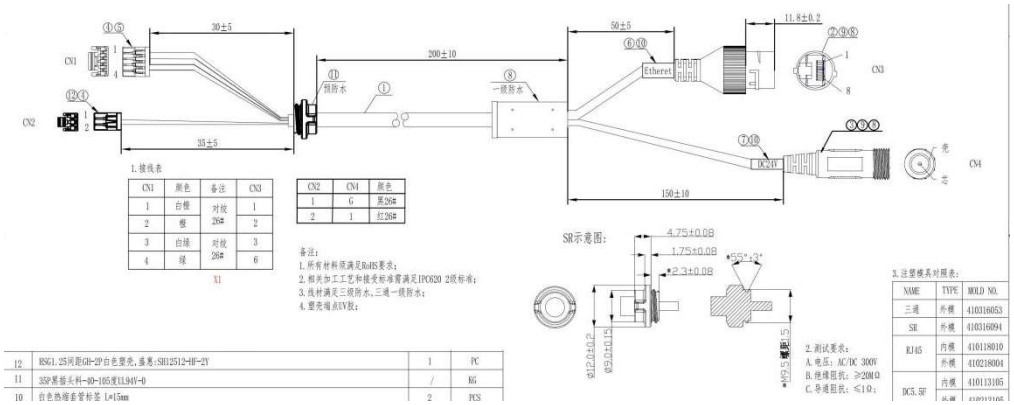


Figure 4 Interface diagram of the camera ports for the S10 series (excluding the 120° version of S10 Lite) (without IO interface)

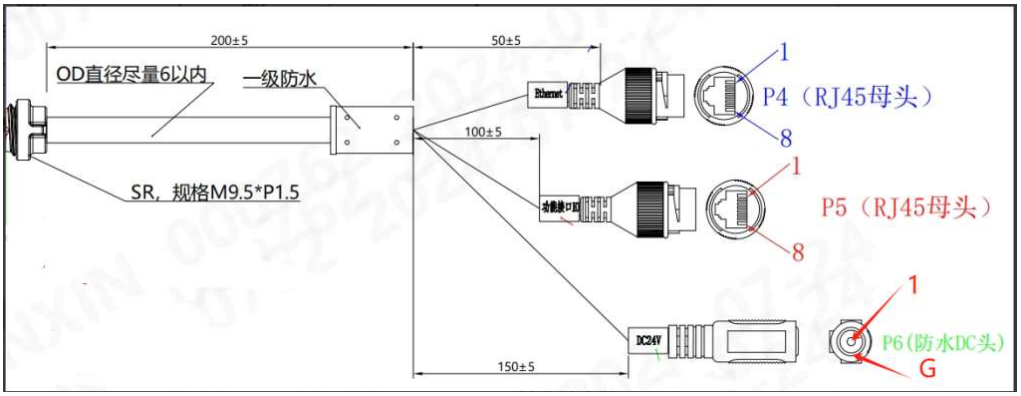


Figure 5 S10 Lite 120° Edition Camera Port Interface Diagram (with IO interface)

2.2.2 IO Wiring Instructions

Port	Attribute	Notes
IN1	Input	Discrete quantity input signal, high active
IN2	Input	Discrete quantity input signal, high active
OUT1	Output	Discrete quantity output signal, ground/open signal
OUT2	Output	Discrete quantity output signal, ground/open signal
COM	GND	Discrete quantity output signal common points, IN1 and IN_COM1 are used in pairs

2.2.3 Cabling Sequence Instructions

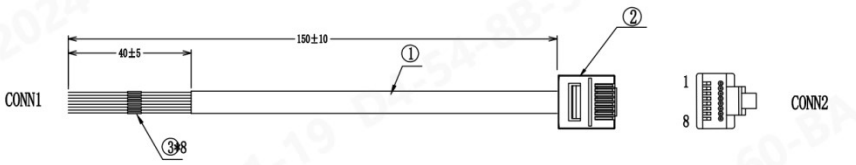


Figure 6 Wiring Sequence Explanation Diagram

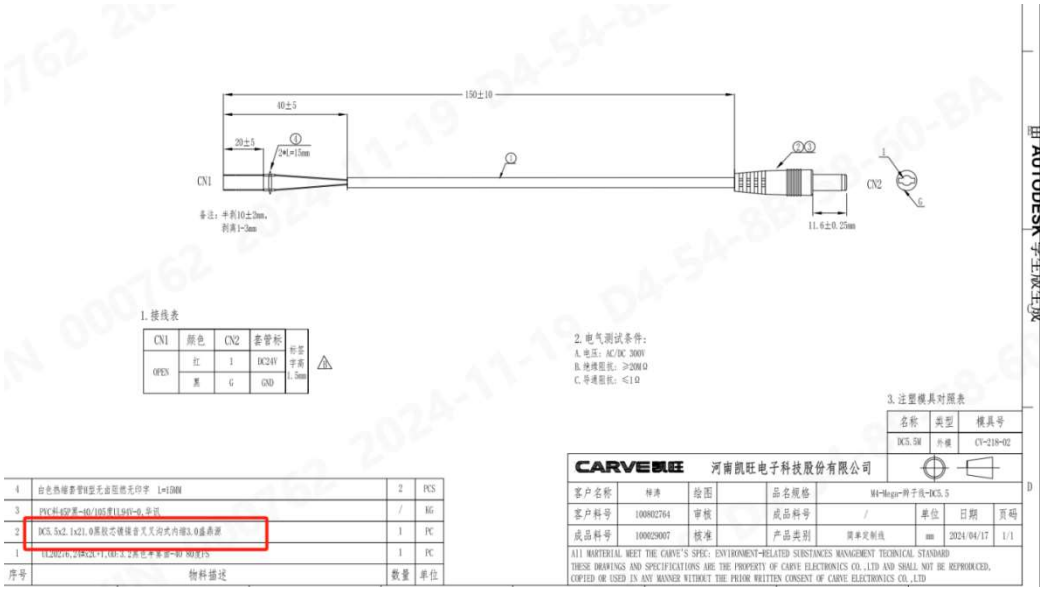


Figure 7 Wiring Sequence Explanation Diagram

IO function port wiring sequence	IO function wiring sequence color	Incorrect wiring sequence definition	Correct wiring sequence definition
1	Orange	OUT_COM2	IN1
2	yellow	OUT_2	IN_COM1
3	Green	OUT_COM1	IN2
4	Purple	IN_COM2	OUT_1
5	Grey	IN2	OUT_COM1
6	Blue	OUT_1	IN_COM2
7	Brown	IN_COM1	OUT_2
8	White	IN1	OUT_COM2

3 Product Information

3.1 Structural Drawings

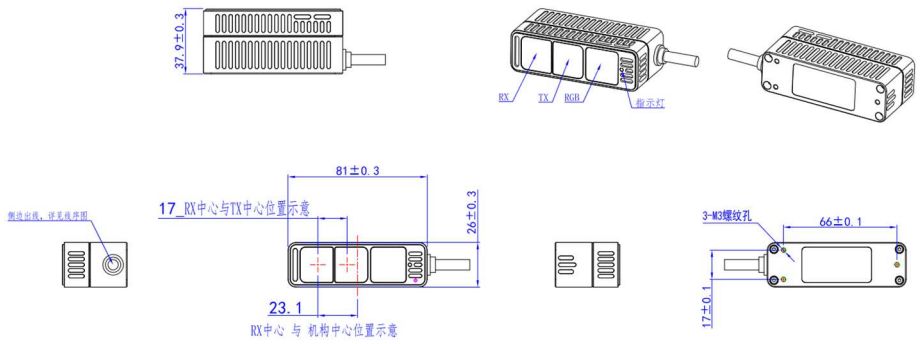


Figure 8 Size Chart of S10 Camera

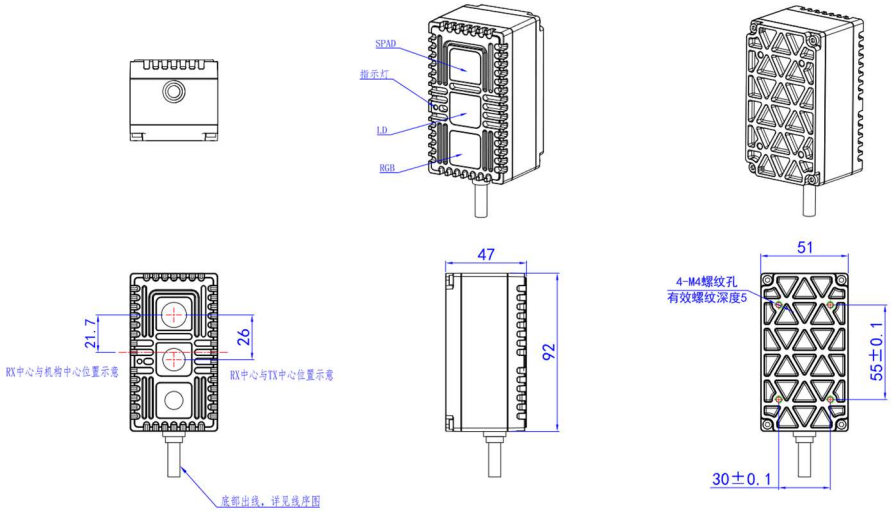


Figure 9 S10Pro Camera Size Diagram

3.2 Camera Components



Figure 11 Introduction Diagram of the Conveyor System Components

Components	Meanings
SPAD	Single-photon avalanche diode

VCSEL	Laser emitter
RGB	RGB module
Power light	Slow flashing after power-on

4 Camera Integration Guide

4.1 Camera Detection Range

4.1.1 3D View of Field of View for S10 Series Cameras

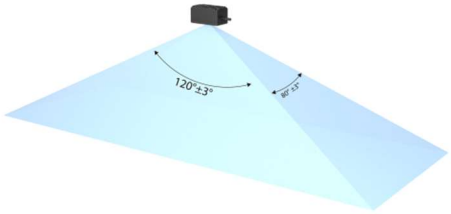


Figure 12 3D view of the S10 series 120° × 80° ± 3° camera field of view

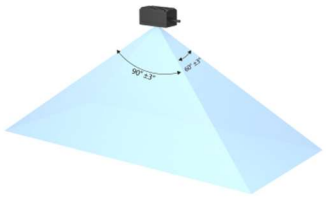


Figure 13 3D view of the field of view of the S10 series 90°×60°±3° camera

4.1.2 2D diagram of field of view for S10 series cameras

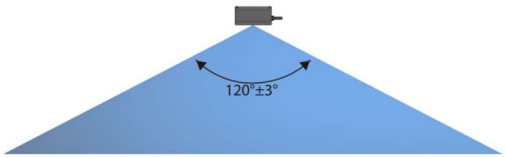


Figure 14 2D view of the field of view of the S10 series camera with an angle of 120° × 80° ± 3°

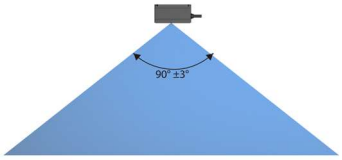
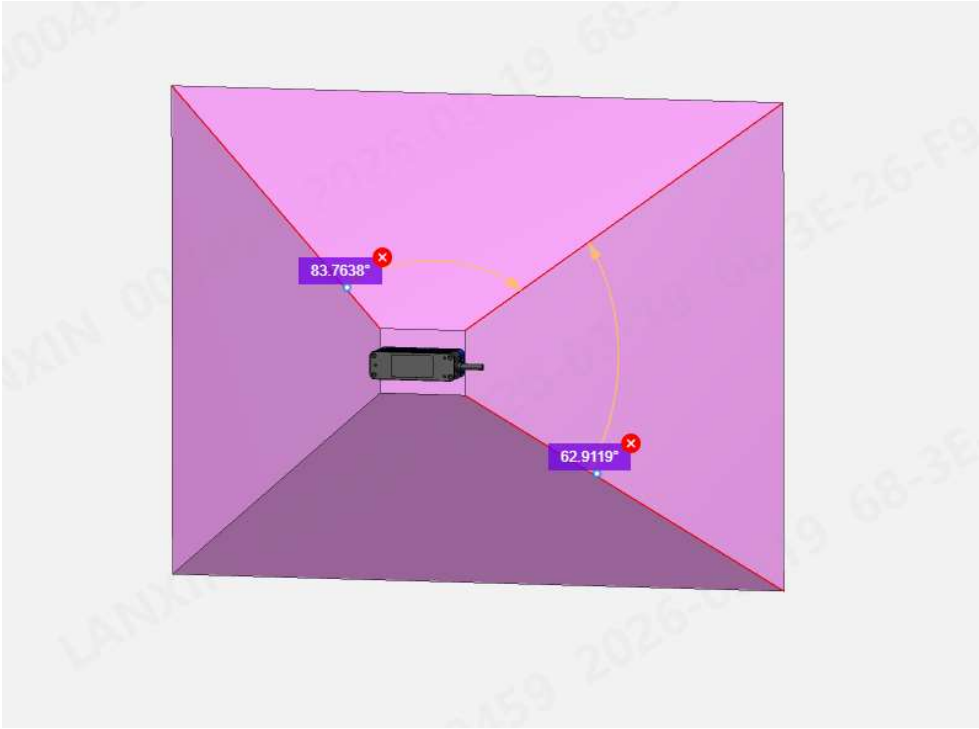


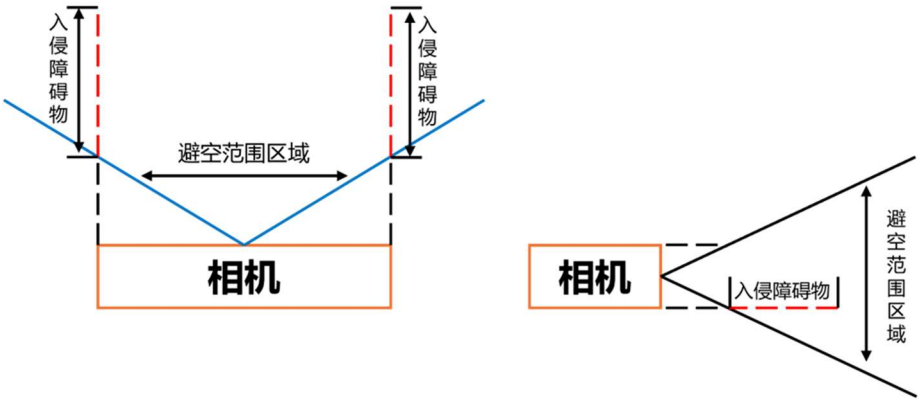
Figure 15 2D view of the field of view of the S10 series 90°×60°±3° camera

4.2 (Important) Installation Notes

4.2.1 Recommended Installation Methods

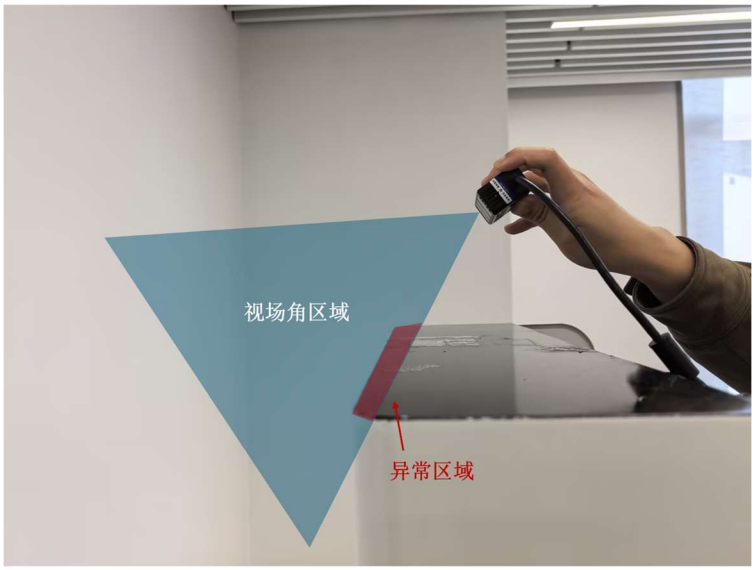
When installing the 3D vision camera, special avoidance measures must be taken for the indicated clearance area (the pink area) in the diagram. No objects are allowed to exist within this area. If any object intrudes into the clearance area, it will directly cause interference to the point cloud data of the equipment, thereby leading to a series of problems such as image noise, incorrect stoppage of obstacles, and abnormal data.





4.2.2 Incorrect Installation Methods

The picture below shows a typical case of incorrect installation. There is an object invading within the camera's blind zone. This installation method directly violates the camera installation standards and will cause interference to the point cloud data. It is the core cause that leads to abnormal noise, false alarms, and incorrect stops of the equipment. On-site installation must strictly avoid such problems.





4.2.3 Reasons for camera malfunction

If the camera is installed in an improper environment, it may interfere with core working principles such as time-of-flight measurement and triangulation, thereby causing a series of usage anomalies. The reasons are as follows:

1、Improper hardware installation: Physical occlusion and field of view (FOV) trimming

Structural frame, ribs, etc. intrude into the sensor's nominal field of view, or are installed eccentrically or with too small an opening, causing the field of view to be trimmed and the sensor to misidentify the structural component as the target. Typical manifestations include fixed blind spots, false close-range distance measurement, and field of view compression, which can lead to device obstacle avoidance missed detection, path misjudgment, and reduced perception reliability.

2、Optical interference: Near-field infrared crosstalk and reflection

This is the most common cause of failure, where the narrow installation space causes abnormal reflection crosstalk of infrared light, mainly in three categories:

First, the inner wall reflection causes multipath interference. Infrared light is received after being reflected multiple times by the inner wall of the structural component, causing the detection time to change and the calculated distance to distort.

Second, strong reflective surfaces cause local failure. Mirror-like reflection occurs in high-gloss smooth materials (polished metals, mirrors), resulting in local signal failure to receive or overexposure saturation.

Third, strong reflection of close-range structures causes high light pollution. When the distance between the sensor and the structural component is less than 10cm and the material is bright and smooth, the strong reflected light causes the receiver to saturate. The iToF camera phase is disrupted by highlights and cannot calculate the flight time. Such disturbances can cause data distortion, accelerate receiver aging, and affect the core functionality of the device.

3、Optical interference: Interference from optical Windows

Interference caused by the material and cleanliness of optical Windows such as the protective glass in front of the sensor is mainly divided into three categories:

First, multiple reflections cause artifact interference. Reflections on the inner and outer surfaces of the window create artifacts that mix with the effective signal, resulting in depth map gimmings, flying spots, and lidar point cloud duplication

Second, contamination and fogging cause a decrease in the signal-to-noise ratio. Closed Spaces can cause fogging and dust accumulation on Windows, making it impossible for sensors to recognize effective signals

Third, mismatched materials cause signal attenuation. The signal may not penetrate ordinary glass or acrylic, resulting in global data loss. Window interference can lead to increased measurement errors, damage to the lens, and increased maintenance costs. Replacing high-transparency glass can improve the situation.

4、Other hidden influencing factors

First, installation stress: too tight snap causes lens micro-deformation, optical axis offset, and ToF camera optical axis offset causes distance deviation.

Second, thermal interference: Poor heat dissipation in confined Spaces, sensor temperature rise, thermal deformation of the window causes distance drift, which is prone to large-scale failure in the long term.

4.3 SDK

Refer to the MRDVS host computer user manual and SDK development guide.

4.4 Operating environment and reliability

Project		Specifications
Working environment	Temperature	-20℃-- 60℃
	Illumination	0KLUX--100KLUX
Storage environment	Temperature	-40℃--85℃
	Humidity	Relative humidity：Max 90% RH
Normal operating temperature rise		Exterior temperature rise <25℃
ESD grade		Contact discharge ±4KV, air discharge ±8KV

RE grade	Compliant with GB 9254 CLASS A specifications
Working life	3 years
Environmental certification	RoHS

5 Quick Usage Guide

5.1 Packing list

No.	Description	Pictures
1	S10 Camera	
2	8PIN function line	
3	2PIN power cord	

5.2 Connection test

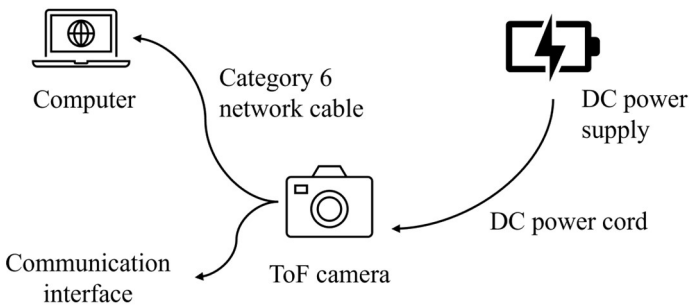


Figure 16 Schematic diagram of connection testing

5.2.1 Camera power supply

- Insert one end of the DC power cord into the camera's DC power interface and connect the other end to a DC power adapter.
- Observe the camera power indicator light. When the indicator light is blue and flashes slowly, it indicates that the power is on normally.
- If the indicator light doesn't light up or shows an anomaly (such as a constantly lit red light), check if the power interface is plugged in and if the power adapter is powered on.

5.2.2 Network configuration

- Connect the camera's communication interface directly to the computer's network port using a Category 6 cable
- The camera comes with a factory default of 192.168.100.82.
- Set your computer's IP to be in the same network segment as your camera and turn off your computer's firewall.

5.2.3 Running the software

- Run the accompanying upper computer software to obtain the camera image, check if the image is clear, without screen flickering or lag, and confirm that the connection is normal.
- Refer to the interface instructions or the LxCameraViewer upper
- computer software user manual for use.
- If you have any other questions, contact Maier Microvision sales or technical support for further assistance.

5.3 Interface Instructions

LxCameraViewer upper computer software supports opening and viewing on multiple devices, supports viewing depth map, intensity map, point cloud map, RGB map output, and supports reading and setting parameters such as basic camera functions and basic algorithm configuration. Users can view and save the image data of the camera, as well as detailed data of the configuration and algorithm within the camera through the software.

The initial interface of this software is divided into three major sections: the list bar (blue area No. 1), the menu bar (green area No. 2), and the image display bar (red

area No. 4). Double-click in the menu bar area to maximize/minimize, and hold down the left mouse button and move the mouse to drag the interface.

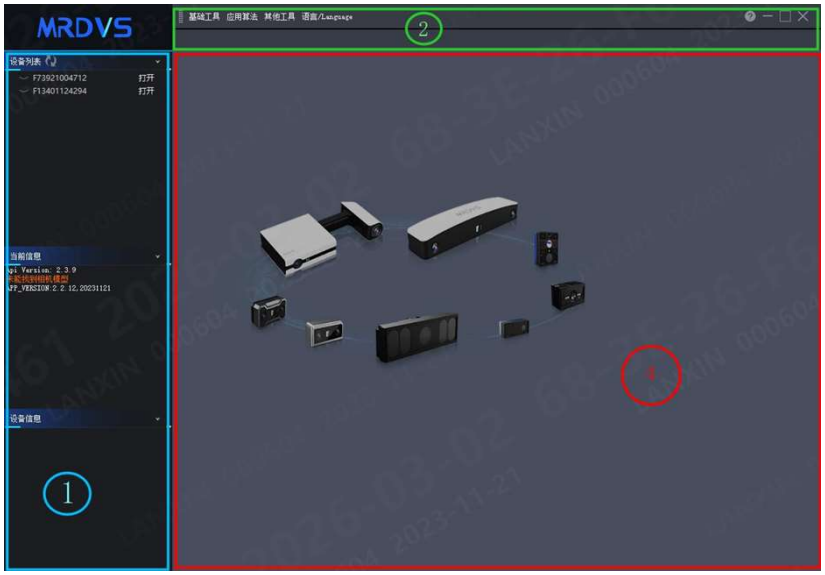


Figure 17 Overview of the software interface (Camera not turned on)

When the camera is turned on, there is an additional function bar on the right side of the interface (area 3), which can be accessed by clicking on the "Basic Tools", "Applied Algorithms", "Others" menu at the top. Among them, the basic Tools offer functions such as image display, 2D Settings, 3D Settings, filtering, etc.

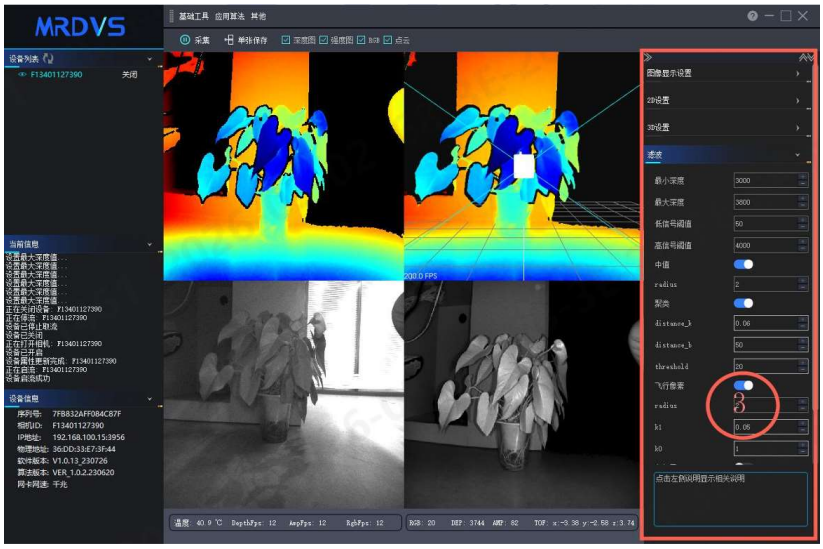


Figure 18 Overview of the software interface (Open the camera)

The device list is located in the top left corner of the interface. Click the text [On] or [Off] to turn the camera on or off. The device list, located in the middle of the left side of the interface, shows camera information, software operations,

operation results, and fetch results. Device information is located in the lower left corner of the interface, showing information about the devices currently selected in the device list.

Clicking the [Capture] button allows the device to start or stop capturing images. Once clicked, the button will be unavailable until the operation is completed. Click the 【Single Save】 button to save the image of the stream data that has been opened. Click the checkboxes of [Depth Map] [Intensity Map] [RGB] [Point Cloud] to open images in the desired format as needed.

For more detailed information, please refer to the LxCameraViewer software user manual.

6 Frequently Asked Questions

No.	Questions	Introductions
1	After opening the camera with the upper computer software, there is no data.	• Please check if the firewall is turned off. It needs to be turned off. • Set the IP address within the same network segment • Check if there is any problem with the switch • Replace the computer • Configure multiple network segments
2	The upper computer has obtained multiple IP addresses.	In the local area network, the same IP has been captured by the camera. You need to select the desired IP to open.
3	Software installation location	If the default C drive is installed, there might be a need for permission settings.
4	The edges of the nearby white object are not very precise.	Optimize the glint algorithm, but there may be some deficiencies in the detection of black objects.
5	The reflective posts are overexposed in a large area.	Activate the glare suppression algorithm
6	The camera software fails to display images or the depth map is very small.	The upper computer software needs to be updated to a version after Lanxin-MRDVS-1.3.66.0606.
7	Failed to obtain exclusive application permissions	Ensure that only one user is connected to the camera at any given time (versions 2.4.50.0720 and later will enter multi-user mode)

8	The imaging of the detected object is sparse or missing.	• Check the high exposure parameters (typically ranging from 600 to 1200) • Check the low signal threshold (generally ranging from 10 to 30)

7 Notes

- This product emits invisible laser during operation. During use, it is necessary to avoid damaging the eyes.
- The laser emitted by this product complies with Class 1 safety level. According to EN60825 requirements, it will not cause harm to the human body during normal use.

More supports

http://www.mrdvs.cn/
https://space.bilibili.com/3546755102672963
https://github.com/Lanxin-MRDVS/CameraSDK
service@mrdvs.com
400-025-6680

WeChat Official Account